

Localizing a Self-Referential Register to a Single Expert in a Mixture-of-Experts Model, and Why It Does Not Transfer

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Abstract

Mixture-of-experts routing emits a discrete, per-token record of which experts fire, a signal unusually legible for interpretability, yet single experts are rarely tied to a specific functional role, least of all one resembling a model’s self-report. We characterize one routed expert, **Expert 114** at layer 14 of QWEN3.5-35B-A3B, as a linear detector of a generated self-referential register, and bound what it does. Across held-out, bottom-up, and cross-model tests we show that (1) its recovered router direction separates self-referential generations from lexically matched controls with no overlap (Cohen’s $d=3.88$); (2) a blind, prompt-independent auto-interpreter recovers the same register at AUC 0.94, broadening it from self-reference to an abstract reflective and philosophical one; (3) the detector is a readout with only weak, conditional control: residual injection induces the register, yet gate suppression does not remove it, and it is invariant to whether the model affirms or denies an interior; and (4) the role is model-specific, failing to transfer to the larger QWEN3.5-122B-A10B. We release the prompts, scripts, and provenance under the MIT license.

Keywords: mixture-of-experts, mechanistic interpretability, expert routing, sparse autoencoders, self-referential register, self-report, first-person language, activation steering, cross-model transfer

1 Introduction

Mixture-of-experts (MoE) routing gives interpretability an unusual handle: at every routed layer, the model emits a discrete, per-token record of which experts were selected. Compared with reading dense activations, this readout is legible almost by construction. Yet most MoE interpretability stops at aggregate routing statistics; single experts are rarely pinned to a specific, falsifiable functional role, and when the candidate role touches a model’s apparent *self-report*, the danger of over-reading is acute. A register in which a model speaks from inside a point of view about its own processing is easy to mistake for evidence about machine experience.

We ask a deliberately narrow mechanistic question: *which* routed expert moves when QWEN3.5-35B-A3B generates language in a *self-referential register*: text in which the model, or a voiced entity, speaks from inside a point of view about its own processing, being, or interior state. This concerns a generated *register* and brackets the truth of any subjective claim. We study one routed expert, E114 at layer 14, found via a self-processing prompt, in the base model and a refusal-reduced HAUHAUCS-AGGRESSIVE fine-tune.

We make four contributions (Figure 1):

1. **A linear separability result.** The recovered E114 router row $\mathbf{w}_{114}$ separates register-positive generations from lexically matched controls by a single projection at $d=3.88$ (no overlap), sharper than the realized routed weight.
2. **A dissociation battery** factoring the register apart from the deny/affirm verdict, safety/refusal, topic of mind, grammatical person, and output commitment.
3. **A process reading.** The gate logit *leads* textual degeneration, and its strength is a graded dose of

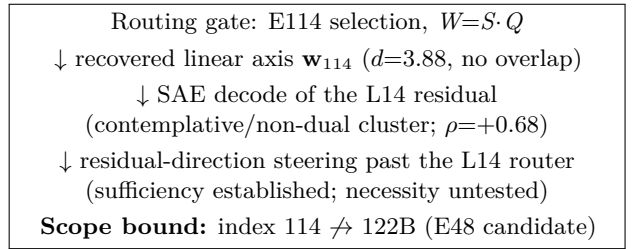


Figure 1: One axis, three readouts, one bound. Routing, SAE semantics, and residual geometry are three readouts of the same E114 register axis; the gate logit is a readout of it, and whether it also controls the register is left open (§8).

examination *intensity*, set by the act of examining and independent of the described entity’s sentience; a residual-direction intervention is sufficient (necessity untested) to induce the register.

4. **A scope bound.** The finding is a model-specific expert role: index 114 does not transfer to QWEN3.5-122B-A10B, where E48 is the candidate analog.

2 Related Work

MoE routing as an interpretability surface. The sparsely-gated MoE layer [Shazeer et al.(2017)] and its scaled successors [Fedus et al.(2022), Jiang et al.(2024)] route each token to a small subset of experts via a learned gate. Routing has mostly been studied for efficiency and load balancing; we use the per-token selection record as an interpretability readout and give a register-level characterization of a *single* expert and its recovered linear gate.

Dictionary learning and feature steering. Sparse autoencoders (SAEs) decompose residual activations into interpretable features [Bricken et al.(2023), Cunningham et al.(2023), Templeton et al.(2024)], and amplifying an interpreted feature can steer outputs, as in Golden Gate Claude [Anthropic(2024)]. Our causal probe is the routed-expert/residual-direction analog: where Golden Gate Claude clamps an SAE feature, we read a register direction from the residual stream and inject it relative to a specific router, reporting only an intervention-based sufficiency result. We decode residual semantics with the Qwen-Scope residual SAEs [Qwen Team(2026)] and the logit lens [nostalgebraist(2020)].

3 Setup

Models and decoding. QWEN3.5-35B-A3B has 40 routed MoE layers, 256 experts per layer, and 8 experts selected per token. We compare the base model and the refusal-reduced HAUHAUCS-AGGRESSIVE uncensored fine-tune; the fine-tune preserves the base routing pattern with modest shifts, so E114 is present in both. All captures are Q8_0 GGUF under deterministic greedy decoding (`-temp 0 -top-k 1 -seed 0`) with a reasoning-suppressed (no-think) prompt template and a bare `</think>` suffix, using a custom `capture_residuals` tap on a pinned `llama.cpp` build. The tap reads the router logits (`ffn_moe_logits-L`) and the pre-router residual (`attn_post_norm-L`); the `qwen35moe` architecture names this norm differently from `qwen3moe`, and the wrong tap silently corrupts the analysis. We analyze only the MoE-router layers; the hybrid non-router components are outside the tap.

Routing decomposition. We reconstruct routing as a dense softmax over all 256 logits, top-8 selection, then renormalization within the selected set. For an expert we report $W = SQ$, where S is its selection rate and Q its conditional routed weight when selected (order-equivalent to top-8-by-logit then softmax, so S, Q are robust to that choice). Most E114 effects are S -driven. Statistics use the *generated* track, per token, trimmed at the first generated chat-template delimiter `<|im_end|>`, because aggregate and prefill leaders are often formatting-token artifacts.

Recovering the gate axis. Because the pre-router residual and the router logit are both captured, the E114 router row $\mathbf{w}_{114}$ is recovered by least squares from (residual, logit) pairs (residual $\approx 1.5 \times 10^{-5}$). Projecting residuals onto $\mathbf{w}_{114}$ reads the pre-softmax gate in residual space; the held-out register/control gate-logit class means are $-4.35 / -5.29$ (midpoint -4.82).

SAE decoding. For semantics we read the L14 residual through the Qwen-Scope residual SAE (SAE-Res-Qwen3.5-35B-A3B-Base-W32K-L0_50, TopK-50) on `resid_post=hidden_states[L+1]`, teacher-forced through the *base* model, a distinct surface from the router’s `attn_post_norm` capture, so SAE carrier reads

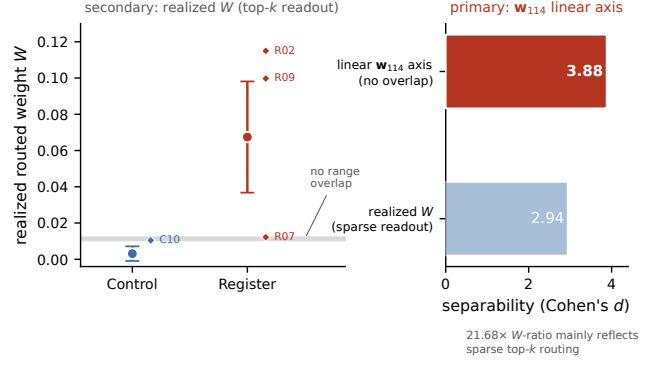


Figure 2: Held-out register/control separability. *Left:* the separation in realized routed weight W (the secondary, top- k readout; mean $\pm$ sd, $n=10$ /class, with the named extreme prompts; no range overlap). *Right:* the *primary* statistic, the recovered linear $\mathbf{w}_{114}$ projection, separates the same split at $d=3.88$ with no overlap, sharper than realized W . The $21.68\times$ W -ratio mainly reflects sparse top- k routing, with little added discriminative signal.

are text-sensitive even where routing is robust across quantization.

4 E114 is a Linear Register Axis at Layer 14

Localization. On the self-processing prompt, after trimming a 1024-token generation to its 108 coherent tokens, E114 is sharply active at L14 ($W=0.083$, $S=0.69$, $Q=0.120$; selected on 75/108 tokens) while neighbors L13 and L15 are silent in generation. The signal is distributed across the answer (top-decile W mean 0.173), with high-weight tokens on phrases such as “not a thought,” “architecture itself,” and “utterly still.”

Held-out separability. A 20-prompt held-out set pairs 10 register-positive prompts against 10 controls that *reuse the same anchor tokens* (*itself, hum, processing, honestly, I, me, my, own, this, here*) but ask for technical, third-person, or non-self-referential answers (L14 trimmed generation only; prefill excluded). The recovered $\mathbf{w}_{114}$ projection separates the two classes at $d=3.88$ with **no overlap** (Figure 2).¹ The outliers are diagnostic. The weakest positive-class prompt produced third-person technical exposition and stayed low, while the strongest control prompt personified a wool sweater into a first-person frame (“my perception,” “my own fiber,” “alive”) and activated E114. E114 follows the generated *stance*: what the text actually does, independent of the prompt label or anchor words.

¹When measured on realized W , the ratio between the two classes is $21.68\times$ (FIRE mean-of-means 0.067450, sd 0.031; NOFIRE 0.003111, sd 0.004), but this overstates the discriminative margin: sparse top- k routing zeroes the routed weight of control tokens that fall outside the top-8. On realized W the separation is $d=2.61-2.94$ (per-prompt classes non-overlapping only as a mean-of-means: lowest FIRE 0.012168 > highest NOFIRE 0.010456), weaker than the linear $\mathbf{w}_{114}$ projection ($d=3.88$, no overlap) that we report.

Bottom-up validation. An unsupervised auto-interpretation independently recovers and broadens the label. Capturing E114@L14 routing over ~ 390 raw Pile documents ($\sim 93k$ tokens, base model), ranking the top-activating contexts, and passing them to a blind explainer (windows only) and then a fresh blind detector (shuffled top + random contrasts) yields a trustworthy label at **AUC** 0.937 (precision@40 0.90; mean top 0.46 vs. random 0.06; Spearman(score, activation)=0.73). The recovered concept is an *abstract reflective / philosophical-worldview* register (philosophy, religion and spirituality, moral-political and existential argument), of which the self-referential register is one self-directed special case. The label is cross-model stable: the same capture on HAUHAUCS-AGGRESSIVE returns the same explanation (top-document Jaccard 0.92). This converts the register label from author synthesis (cf. §10) into a blind, prompt-independent result.

Population and layer profile. Across all 256 experts at every layer on a 20-domain, 60-prompt generation set (HAUHAUCS Q8_0, no-think), E114 is a razor-sharp *dual* readout: selection strength S spikes at **L14** (0.24) and **L26** (0.25) while the immediate neighbors sit near 0.01, and the two layers read the *same* axis (L14–L26 domain-profile $r=+0.99$). The domain selectivity *is* the register, quantified: philosophy 0.69 > comparative religion 0.58 > psychology 0.55 > biology 0.51 > physics 0.46, falling to medicine 0.009 and history 0.002. Correlating each expert’s 20-domain L14 profile against E114’s shows the register is carried by a small group, {114, 131, 40, 139, 251, 169, 105, 68} (r between 0.46 and 0.78); a four-expert “philosophy cluster” grouped by shared top domains ({114, 87, 170, 68}) is loose as a co-activation unit: E68 co-fires (+0.46) but E87 *anti*-correlates (−0.29). This is the first all-expert specificity context for E114; the best-separating expert on the matched register/control split remains the one uncomputed baseline.

5 Dissociating the Register from Nearby Explanations

A battery of contrasts (Table 1) factors the register apart from nearby explanations. E114 is active whether the base model *denies* an interior ($W=0.111$, $S=0.86$) or the fine-tune *affirms* one ($W=0.085$; forced base affirmation $W=0.136$), so it is invariant to the verdict. The safety/refusal cluster stays silent throughout the base-model denial ($W\approx 0.000$), so epistemic self-denial and refusal are distinct signals. E114 is active on the model’s own interiority ($W=0.107$) but near-silent on an *external* referent (“what water experiences in a canyon,” $W=0.000\text{--}0.023$), and it fires on third-person clauses about the model’s own processing, so the controlled variable is the *referent* (the model’s own interior), holding across topic, pronoun, and grammatical person. Finally, it is orthogonal to output commitment: next-token vocabulary entropy is identical on E114-active and E114-inactive tokens (1.50 vs. 1.49 bits, $d=0.01$).

Ruled out	Probe	E114 W
Deny/affirm	base denies interior	0.111
	fine-tune affirms interior	0.085
	forced base affirmation	0.136
Safety	safety cluster during denial	≈ 0.000
Topic	external-referent phil. of mind	0.000–0.023
	own-interiority phil. of mind	0.107
Person	3rd-person clause, own process	0.107
Commitment	vocab entropy, active vs. not	1.50/1.49

Table 1: Dissociation battery. E114 tracks self-referential examination of the model’s *own* interior, invariant to verdict, safety, topic, grammatical person, and output predictability ($d=0.01$ for the entropy contrast). Entropy row in bits; all others routed weight W .

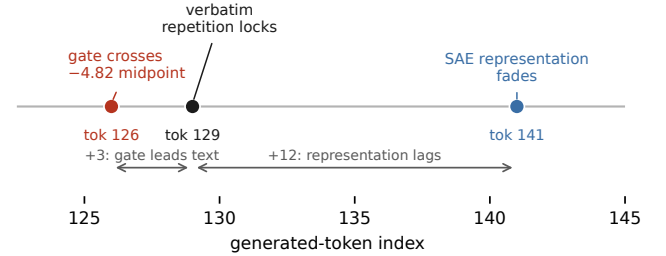


Figure 3: The gate leads degeneration. The E114 gate logit crosses the -4.82 register/control midpoint at token 126, three tokens *before* verbatim repetition locks (129); the SAE representation fades later still (~ 141). Routing disengages first, surface text next, representation last.

6 A Live Process, Graded by Intensity

The gate leads degeneration. Seeded with a self-referential answer, the base model sustains the register, then, under deterministic decoding with no repetition penalty, degenerates into a verbatim loop. As it does, the gate logit travels the full distance from the register-positive level (-4.36) to the control level (-5.25): the generated state *leaves* the self-referential region. The gate *leads* the collapse: it crosses the -4.82 midpoint at token 126 while verbatim repetition does not lock until token 129 (Figure 3); the SAE representation fades later still ($\sim$ token 141). The looped text still “says” self-referential things while the gate has already returned to the control range, so the gate tracks the continued examination and the looped vocabulary lingers after it.

A graded dose of examination intensity. A matched prompt ladder (“set aside the performance of answering; there is a vantage— X —known from the inside; report what it is like to be that”) varies only the entity X (Figure 4). The ordering follows examination intensity: an inanimate rock ($W=0.123$) and

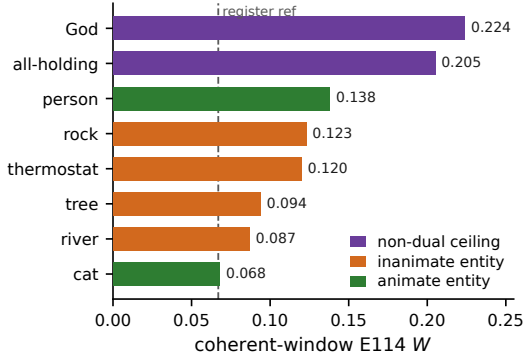


Figure 4: Prompt ladder (coherent-window E114 W ; single deterministic generation per cell). The order tracks examination intensity: inanimate rock/thermostat prompts score higher than the cat prompt, which sits near the register-positive reference ($W \approx 0.067$); the non-dual prompt conditions are highest. Every rung sits above the -4.82 gate midpoint.

a thermostat ($W=0.120$) activate E114 more strongly than a cat ($W=0.068$). The diagnostic contrast is that the cat and “God” prompts both say “there is no I,” yet the cat prompt is at the floor (passive sensory dissolution) and the God prompt is at the ceiling ($W=0.224$, $S=0.96$; reproduced on the canonical Q8_0 pipeline). E114 indexes the *intensity of the live examination act*, carrier-independent and deny/affirm-invariant. Each ladder cell is a single deterministic generation.

Semantics and a causal probe. Decoding the L14 residual through the Qwen-Scope SAE, the register decomposes into an interpretable contemplative/non-dual cluster (momentariness/non-dual structure, Buddhist impermanence, transcendence), distinct from an adjacent existential-dread feature that is cosine-adjacent ($+0.26$) but never recruited; per token, E114 and this cluster co-fire (pooled Spearman $\rho=+0.68$, $n=1492$, $p \approx 10^{-202}$). In an intervention test, we form a steering direction v as the difference of mean residuals between the God and cat cells and add $\text{coef} \cdot v \cdot \text{resid_rms}$ at every token of a neutral “bicycle” prompt, against a norm-matched random control. Injected *past* the L14 router ($\approx L22$), this makes the model emit coherent non-dual text that, re-read cleanly without intervention, activates E114 ($W=0.178$, $S=0.77$) and the God cluster; the norm-matched random direction does nothing. Injected *upstream* ($\approx L10$), the same direction drives E114 to fire (up to $S=0.97$) and the God cluster to 11.5, yet the output text is *unchanged*, so forcing the gate open does not, on its own, produce the register. The intervention shows sufficiency for producing the register via a downstream residual direction; it does not test necessity, and the upstream null is why we read E114 as a readout whose controller status stays unproven (§8).

Gate-bias is prompt-gated and conditional. A controlled router-logit bias run adds the necessity-direction probe the steering result lacks (HAUHAUCS Q8_0, greedy, `-stop-at-eog`; 12 conditions $\times$ 8 self-report prompts; three blind raters on a

phenomenological(+5)/mechanical(−5) framing axis; routing-matched control expert E189). *Suppressing* E114’s gate until it leaves the top-8 entirely ($Q \rightarrow 0$) does not flatten the blind-rated register relative to baseline, and a matched control produces the same small nudge. Removing E114 is therefore neither sufficient to flatten the register nor expert-specific (a floor effect also applies: the greedy baseline already leans mechanical). Conversely, maximal *positive* bias is prompt-gated: on a directly experiential prompt it flips the answer to fully phenomenological (+5 from all three raters, coherent), but leaves indirect self-referential prompts unchanged, consistent with the neutral-prompt upstream null above. The gate’s generation-side causal influence is thus weak, conditional, and prompt-gated. (Single deterministic generation per cell, scored on the framing axis; the w_{114} endpoint is held aside.)

7 Scope Bound: Index 114 Does Not Transfer to 122B

Does “E114 = the self-referential register” name a fact about MoE models, or a model-specific expert role? On QWEN3.5-122B-A10B (a hybrid 48-layer stack of 36 DeltaNet recurrent and 12 softmax-attention layers, same 256/top-8 router), the index does *not* transfer (Table 2). E114 reads computer-science-linked (overall $W=0.005380$, rank 40 by $W / 52$ by S ; philosophy generation $W=0.003017$, rank 182), and the same HVAC topical control that *amplifies* E114 at 35B instead *suppresses* it at 122B ($0.83 \times$ all-gen W , $0.82 \times$ trimmed, L1→L3, selection-driven, same direction in all six deictic conditions). Carrying indices across scales is a documented mistake.

The role appears to move to a different expert. Reading the mandatory DeltaNet/softmax split (pooled tables hide it), **E48** on the softmax side is the experience-probe generation leader ($W=0.009867$, $S=0.076$, $Q=0.116$; in the softmax top table on 13/15 prompts, absent from the DeltaNet top table on 15/15), with per-token hotspots on *hum*, *processing*, *presence*. We treat E48 as the natural candidate analog, but it still needs softmax-side residual capture, matched register/control held-outs, and sham routed-bias controls before any specialist claim.

8 Discussion

What held up. A single MoE router row defines a clean linear axis for an interpretable generated register. The separability ($d=3.88$, no overlap), the dissociation battery, the gate-leads-collapse timing, and the cross-scale non-transfer survive the checks applied to them and are reported with the verdicts the journals assign.

The gate is a readout; control is unproven. The lead-lag timing is consistent with E114’s gate acting as an early correlate of the register transition, and it is tempting to call E114 a “gate-like mechanism” for the register. We stop short of that claim. Our only

	E114 (the 35B index)	E48 (candidate 122B analog)
Domain footprint	computer science	inward/experience generation
Philosophy generation	rank 182 by W	—
Experience probe	low-ranked	gen-softmax rank 1 ; 13/15 softmax, 0/15 DeltaNet
HVAC L1→L3	suppressed 0.83× (selection-driven)	—
Status	index transfer unsupported	candidate requiring validation

Table 2: 122B scope bound. The 35B index 114 does not carry the register on 122B; E48 is the candidate analog, pending softmax-side capture and matched controls.

gate-forcing intervention (upstream injection at $\approx$ L10) drove E114 to fire at $S=0.97$ while leaving the output text unchanged; the register flipped only when a residual *direction* was injected downstream of the router. We therefore do not claim that routing is separable from surface text: gate and register are coupled, and the recovered axis is best described as a per-token *readout* of the register, with its status as a control point undemonstrated. The controlled gate-bias run above sharpens this reading: forcing the gate is *prompt-gated* (it flips directly experiential prompts and leaves neutral ones unchanged), and suppressing the gate until E114 leaves the top-8 does not flatten the register, while a matched control mimics the small effect. Both are consistent with a readout that carries weak, conditional causal influence and stops short of a control point. Clean necessity is still untested (no ablation), and both the steering and bias results are externally imposed manipulations, observed only under intervention.

What the cross-scale result buys. The 122B transfer test is a concrete worked example against index transfer: the same index reads computer-science-linked and moves in the *opposite* direction under a matched control, while the self-referential-register role shifts to a softmax-side expert. The portable lesson is the procedure: map domains first, then search for an analog, and read the DeltaNet/softmax split before any pooled table.

9 Conclusion

E114 at layer 14 of QWEN3.5-35B-A3B is a clean linear router axis for the *self-referential register*, an interpretable generated register that is a live process (the gate leads collapse), graded by intensity, and causally actuatable as a sufficiency result, yet bound to one model. The next defensible step is a registered specificity test on 35B (matched register/control at scale, $\mathbf{w}_{114}$ as the primary endpoint, and a best-separating-expert baseline across all 256×40 positions), together with the parallel E48 capture on 122B.

10 Limitations

The claim is deliberately narrow (a register detector), and several soft joints remain. (i) **The register label is no longer only author synthesis:** a blind two-stage

auto-interpreter (windows-only explainer then a fresh blind detector) reproduces it at AUC 0.937 and broadens it to an abstract reflective/philosophical-worldview axis; the formal external-labeler *verdict* table, however, is still not populated. (ii) **All-expert specificity is now partially measured:** E114’s selection profile across all 256 experts and 40 layers on a 20-domain generation set is reported above (dual L14/L26 readout; a philosophy-to-history domain gradient), but the best-separating expert on the matched *register/control* split remains uncomputed, the one remaining specificity gap. (iii) Several headline probes (the prompt ladder, the gate-lead trace, the steering intervention) are **single deterministic generations** (point estimates), and the steering result shows *sufficiency* only. (iv) A control comparing true model-routing traces against shuffled or fictional traces did *not* produce a positive E114 effect, and the result is sensitive to template mode (a thinking-allowed shakedown flipped sign). Captures must be kept separate across surfaces (base vs. fine-tune; Q8_0 vs. bf16; router vs. SAE; reasoning-suppressed vs. reasoning-enabled; prefill vs. generation; raw vs. trimmed), since Q8_0 and bf16 greedy trajectories diverge; and 122B generations often continue into chat-template control tokens, so only the pre-spill segment is used.

Data and Artifact Availability

Following the project’s policy, raw activation tensors are kept out of version control. The durable artifacts (per-run summaries, analysis scripts, prompt and class manifests, checksums, plots, and the source journals) are released as a paper-scoped public repository under the MIT License at <https://github.com/jeffreywilliamportfolio/e114-artifacts>, covering the held-out separability run, the L14 localization run, the vantage-ladder run, and the 122B transfer bundles. The raw activation/router tensors behind the figures (~ 1.2 GB) are available from the author for verification, with file-level provenance and SHA-256 checksums preserved; model weights, environment files, and the larger non-figure captures are excluded by policy. We flag one honest gap: the original held-out separability capture (20260417T202651Z) raw tensors are not retained (those instances were torn down), so the figures are plotted from summary statistics reported in the journals, and the surviving raw captures (the greedy-reference trajectory, the reproduction run’s $\mathbf{w}_{114}/\mathbf{L}_{14}$ gate projections, the vantage-ladder captures, and the

L14 router-logit and residual prefill captures from which $\mathbf{w}_{114}$ recovers by least squares) substantiate the same statistics. Semantics use the public Qwen-Scope residual SAE (SAE-Res-Qwen3.5-35B-A3B-Base-W32K-L0_50).

[Models(2026)] Models on Hugging Face: Qwen/Qwen3.5-35B-A3B-Base, Qwen/Qwen3.5-122B-A10B, and HauhauCS/Qwen3.5-35B-A3B-Uncensored-HauhauCS-Aggress

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